CT-3
Industrial and Operations Management
Time: 30 Minutes

Q1: (5 Marks)
A small garment factory produced 1,200 shirts in one 8-hour workday.
The following resources were used during the day:

Labor: 10 workers, each paid ৳600 per day
Machine time: 6 sewing machines operated for 8 hours, costing ৳120 per machine-hour
Materials: Fabric and accessories costing ৳90,000
Energy & other overhead: ৳6,000

Required:
a) Calculate the Single-Factor Productivity of Labor, expressed as: shirts per worker& shirts per labor-hour

b) Calculate the Single-Factor Productivity of Machines, expressed as: shirts per machine-hour

c) Calculate the Multifactor Productivity (MFP), considering labor cost, machine cost, materials, and overhead as inputs.
d) If the factory improves its process and produces 1,320 shirts the next day using the same inputs, determine the percentage increase in Multifactor Productivity.

☑ Note:

Show all steps clearly.
Use appropriate productivity formulas.
Assume all workers and machines worked the entire day.

Q2(5Marks)
A manufacturing company uses a raw material at a non-uniform rate.

Annual demand is 18,000 units
Ordering cost per order = ৳1,200
Holding cost = 20% of unit purchase price per year
Unit purchase price = ৳50

Supplier Discount Policy
For order quantity less than 1,000 units, unit price = ৳50
For order quantity 1,000 units or more, unit price = ৳48

Production Consumption Overlap

The material is consumed gradually at 75 units/day
The supplier delivers material at a rate of 300 units/day
The firm operates 300 days per year
Shortage Allowed (with Penalty)
Shortage cost = ৳4 per unit per year
Backlogging is allowed

Required:
a) Determine the optimal order quantity that minimizes total annual cost.